SAHANA ARULANANDAM

Bengaluru, IN 560074 | sahana.arul@outlook.com | +91 8390080709

SUMMARY

I am an enthusiastic and creative robotics and mechatronics engineering student, currently pursuing my degree at Christ University, Bengaluru. I am driven by a passion for autonomous systems, robot simulation, and intelligent IoT applications. With hands-on experience in ROS 2 navigation, AMR simulation, swarm robotics, and digital twin development, I excel at tackling cross-disciplinary challenges. As a team leader and researcher, I bring strong technical depth across embedded systems, computer vision, and software development. I am eager to join an innovative, collaborative team where I can leverage my technical expertise and creative problem-solving skills to advance robotics and automation technologies, contributing to projects that will help redefine the future of intelligent autonomous systems and human-robot interaction.

TECHNICAL SKILLS

Robotics & Simulation

- **Frameworks:** ROS 2 (Jazzy), Gazebo Harmonic
- **Navigation:** Differential Drive & AMR Navigation, Path Planning, Obstacle Avoidance, LiDAR, SLAM, PID Control, PWM Motor control, URDFs
- **Sensing:** Sensor Integration (LIDAR, IMU, Ultrasonic, Encoders)
- **3D Modelling:** Blender
- **Industrial Robots:** KUKA KR6 R900, Ati Motors Sherpa RP, Botsync Copernicus.

IoT & Hardware

- **Microcontrollers:** Raspberry Pi, Nvidia Jetson Orin Nano, ESP8266 / ESP32, Arduino Uno / Mega / Nano
- **Protocols:** MQTT, UDP, TCP, ESP-NOW, CoAP Protocols, ThingSpeak Cloud Integration, Blynk IoT, Eclipse Mosquitto

Programming:

- Python, C++, C, Java, Linux, Ubuntu, Git, MATLAB & Simulink

Languages:

- English, Hindi, Tamil

EDUCATION

- **Christ University, Bengaluru** *2023 - 2027*
  B.Tech in Robotics and Mechatronics Engineering
- **Presentation Convent School, Nerul** *2019 - 2023*
  ICSE & ISC Classes 10th and 12th

PUBLICATIONS

- **Conference Paper** “Comparative Evaluation of DWB, MPPI, and RPP Local Planners for Autonomous Mobile Robots in Indoor Environments” *DSATM AMISE 2026*
- **Conference Paper "**Design and Implementation of Hand-Gesture Controlled Robotic Arm using Smart Gloves and Machine Learning**"** *DSI-DMMES 2026*
- **Conference Poster "**Hybrid Communication and Digital Twin Integration for Autonomous Mobility**"** *ICVTTS 2025*

WORK EXPERIENCE.

Intern June 2026 – November 2026
Artificial Intelligence and Robotics Laboratory, IISc Bangalore

- Developing ROS 2 navigation software integrating perception, localization, and map-based planning to integrate OSM with real-time perception.
- Designing and evaluating real-time navigation algorithms for autonomous mobile robots.

Undergraduate Research Intern April 2024 – June 2024
Christ (Deemed To Be University), Centre of Excellence in E-Mobility, Bengaluru, India

- Developed a real-time digital twin of an electric two-wheeler using Unreal Engine 5.
- Integrated ESP8266 sensors, UDP communication, and ThingSpeak cloud monitoring.

PROJECTS

Autonomous Mobile Robot (AMR) Navigation System

- Built a ROS 2 Jazzy navigation stack on Jetson Orin Nano using Gazebo Harmonic.
- Evaluated DWB, MPPI, and RPP planners in simulation and on hardware.

Collaborative Swarm Robotics System

- Developed MQTT-based communication for decentralized multi-robot coordination.
- Implemented PID-based collision avoidance and task allocation.

LEADERSHIP ROLES

Founding Chair - IEEE Robotics and Automation Society (RAS), Christ University | *2025 – 2026*

- Founded and led the IEEE RAS Student Branch.
- Organized robotics workshops, technical events, and industry sessions.

Department Head - Magnovite 2025, Christ University | *December 2025*

- Led departmental planning and event execution.
- Coordinated logistics and student teams.

CERTIFICATIONS

NPTEL Swayam - Advanced Robotics Applications
MathWorks - MATLAB & Simulink Onramp, MATLAB & Simulink Fundamentals